FACTOR INVESTING IN COMMODITY FUTURES

Six commodity factors, each fragile on its own. Combined, a third of the market's drawdown.

Sharpe ratios, net of costs · 26 US futures · 2002 to 2026



-14%

MAX DRAWDOWN

vs -39% for the commodity market, -39% to -66% for single factors

0.12

CORRELATION

with the commodity market: returns largely independent of commodity prices

t = 3.91

SIGNIFICANCE

clears the t > 3 hurdle proposed for new factors (Harvey, Liu & Zhu)

Navon Shapurkar

Quantitative research · backtest on free public data

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Higher Sharpe, a third of the drawdown

An equal-risk blend of six long-short commodity factors delivered a Sharpe ratio of 0.71 (t = 3.91) versus 0.63 for the commodity market, with a maximum drawdown of -14% versus -39%. Correlation with the market: 0.12.



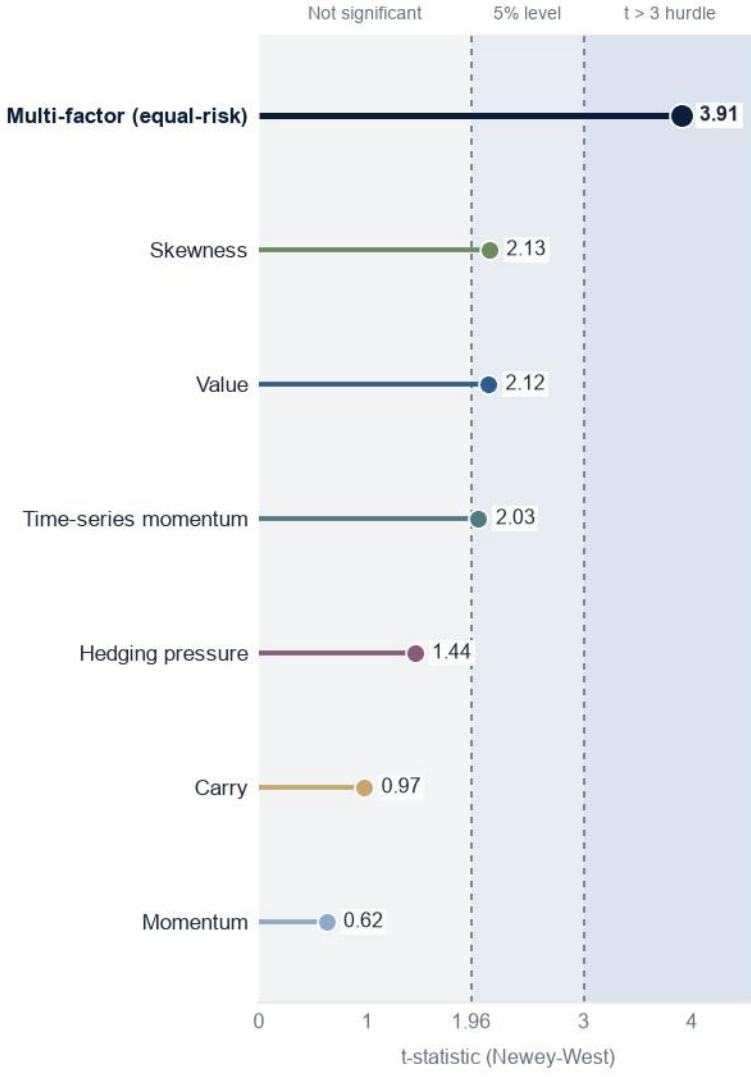
	Ann. return	Volatility	Sharpe	Max drawdown
Multi-factor (equal-risk)	6.0%	8.4%	0.71	-14%
Commodity market (equal-weight)	8.5%	13.6%	0.63	-39%



Data: Yahoo Finance front-month futures, EIA, CFTC Commitments of Traders, FRED, AQR Data Library. 26 US commodity futures, Feb 2002 to Sep 2026. Excess returns net of 5 bps per unit of turnover, including roll costs. Backtested results; past performance is not indicative of future results.

Individually fragile, jointly significant

Three factors pass the conventional 5% test (t > 1.96), but none survives adjustment for testing six factors at once. The pre-specified combination clears the stricter t > 3 hurdle.



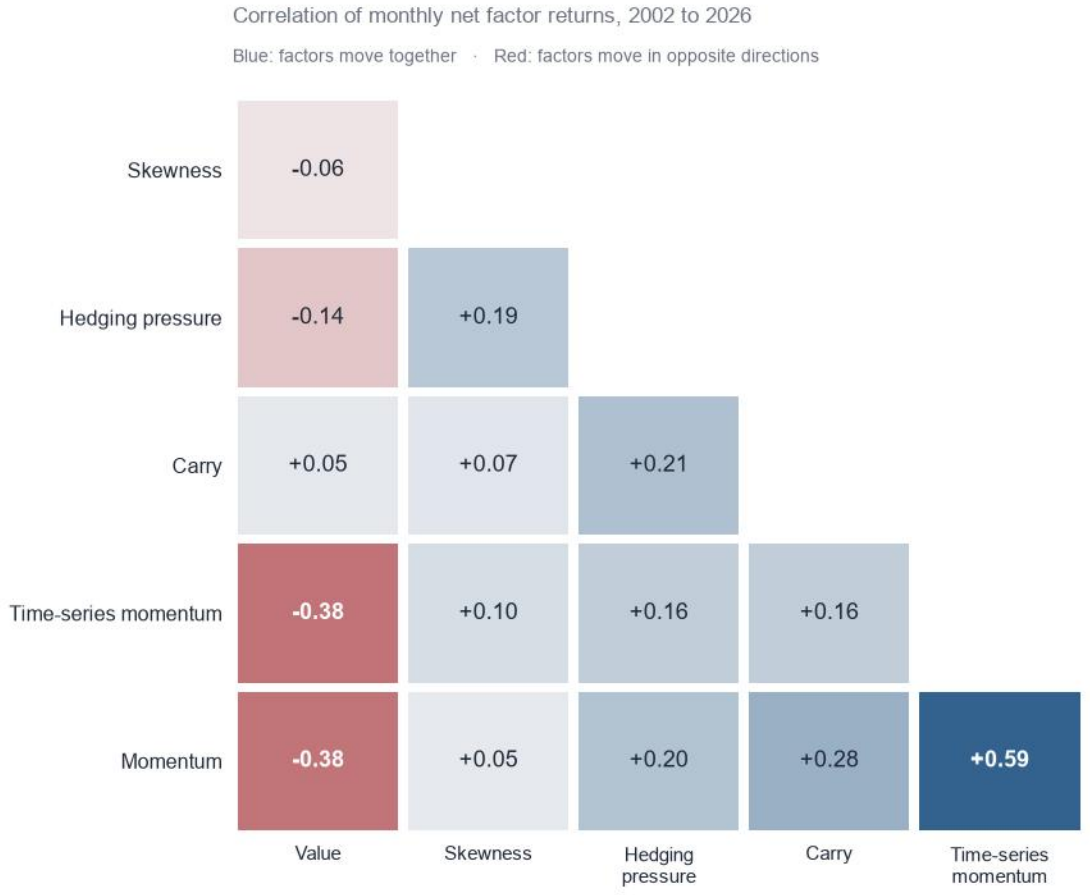
	Sharpe ratio	Max drawdown	Unique α (t-stat)
Multi-factor (equal-risk)	**0.71**	**-14%**	**–**
Skewness	0.46	-47%	0.80
Value	0.44	-40%	2.87
Time-series momentum	0.40	-39%	1.14
Hedging pressure	0.34	-51%	2.02
Carry	0.17	-66%	-0.60
Momentum	0.11	-66%	0.47

Unique α (t): t-statistic of the factor's alpha after controlling for the other five factors and the commodity market (common sample from Nov 2005). Dashed lines mark t = 1.96 and t = 3.

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Why combining works: low correlations

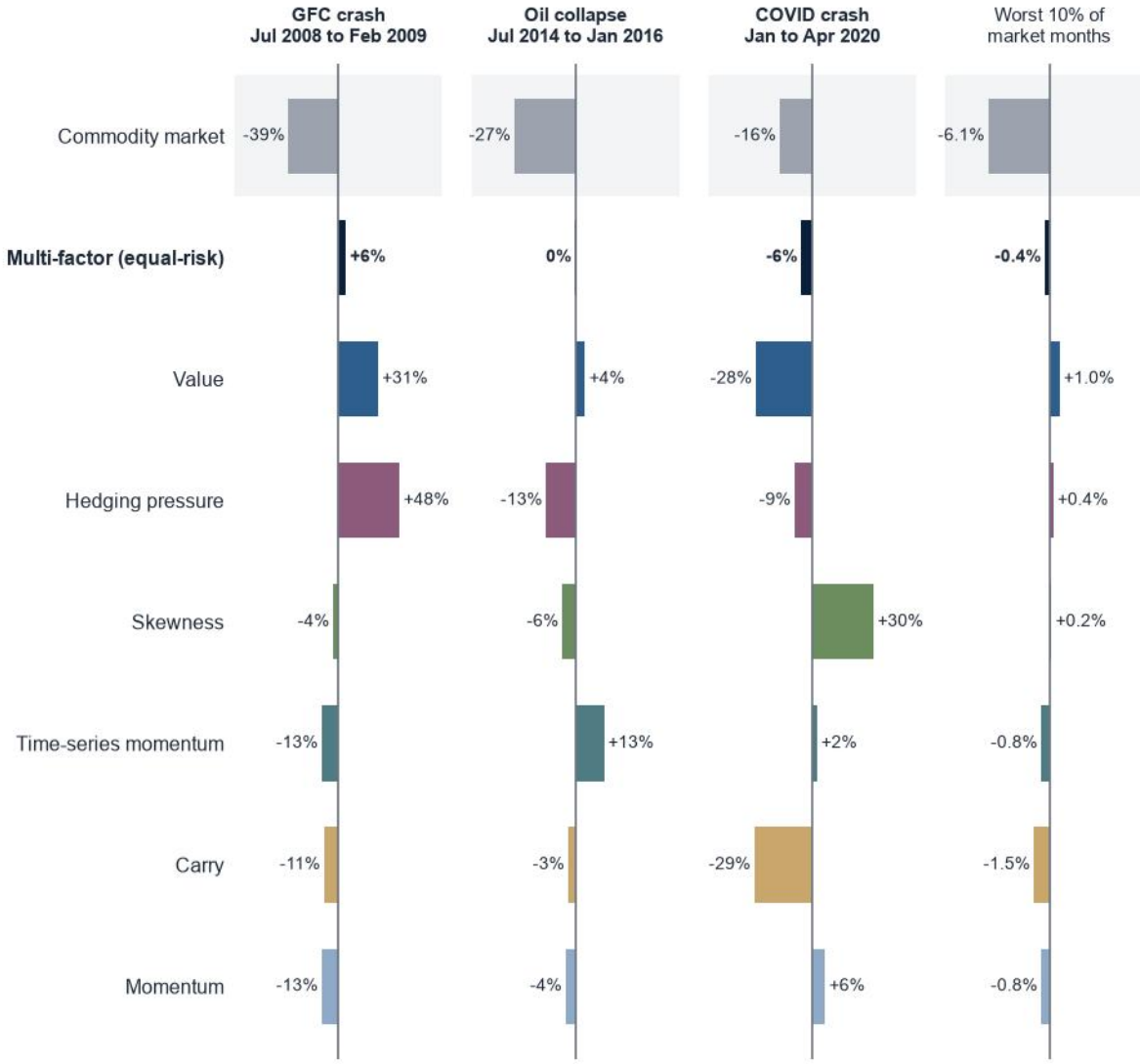
Most factor pairs are close to uncorrelated. Value moves against momentum and trend (-0.38), so its good years tend to offset theirs. Combining six factors cut volatility from 17% to 8%.



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Close to flat while the market fell

In three major sell-offs the multi-factor portfolio returned between -6% and +6%, while the commodity market fell 16% to 39%. Value and hedging pressure gained in 2008; carry lost in sudden crashes.



Cumulative net return over each episode. Last panel: average monthly return in the market's worst 10% of months (combination from Feb 2004).

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Factors depend on regimes; the combination barely does

Individual factors swing by up to 22 points a year between regimes (only 1 of 36 differences reaches |t| ≥ 1.96). The multi-factor portfolio stays within -1.2 to +6.3 points: the factors' tendencies offset.

	Rising vs falling market	High vs low volatility	CPI > 3% vs ≤ 3%	USD up vs down	VIX > 25 vs ≤ 25	Recession vs expansion*
Multi-factor (equal-risk)	-1.2 t = -0.3	+0.5 t = 0.1	+6.3 t = 1.6	+1.1 t = 0.3	-0.6 t = -0.1	-0.4 t = -0.0
Value	-5.8 t = -0.6	**+18.2** t = 2.3	+8.0 t = 0.9	-1.1 t = -0.1	+10.6 t = 0.8	-8.5 t = -0.4
Hedging pressure	-15.1 t = -2.0	+7.0 t = 1.0	+11.9 t = 1.6	+9.3 t = 1.4	+1.8 t = 0.2	+22.4 t = 1.6
Skewness	-8.1 t = -0.9	-3.0 t = -0.4	-0.6 t = -0.1	+6.5 t = 0.9	+3.4 t = 0.3	-12.5 t = -0.8
Time-series momentum	+10.7 t = 1.9	-4.1 t = -0.8	+0.7 t = 0.1	+0.1 t = 0.0	-7.9 t = -1.0	-0.6 t = -0.0
Carry	+2.3 t = 0.3	+2.3 t = 0.3	+12.6 t = 1.7	-4.2 t = -0.6	-11.0 t = -1.2	-16.6 t = -0.9
Momentum	+11.0 t = 1.3	-7.1 t = -0.9	+7.4 t = 0.9	-3.5 t = -0.5	-11.4 t = -1.0	-2.0 t = -0.2

Cell value: difference in annualised net return between the two states (percentage points); outlined where |t| ≥ 1.96 (Welch, not adjusted for autocorrelation). Regimes use only information available before each month, except *recession (NBER, dated ex post; 20 months).

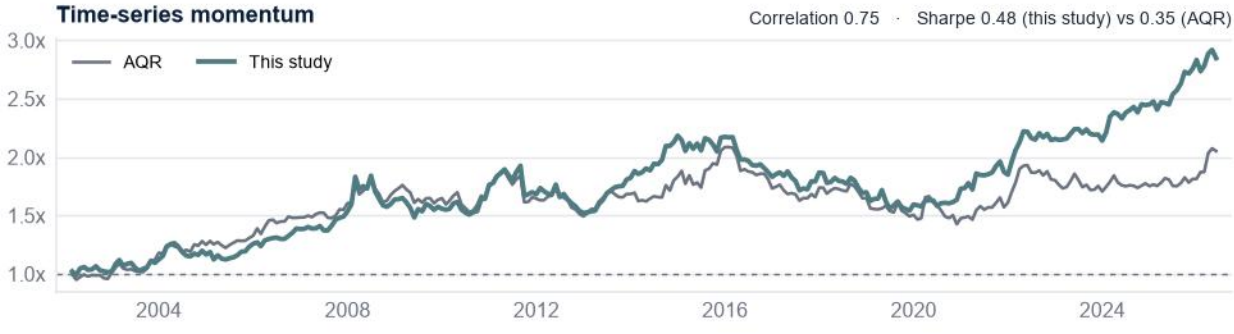
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Independent check: our factors track AQR's

Correlations of 0.75 to 0.79 with AQR's published commodity factors confirm the replication. Our Sharpe ratios run higher, so treat absolute performance as optimistic.







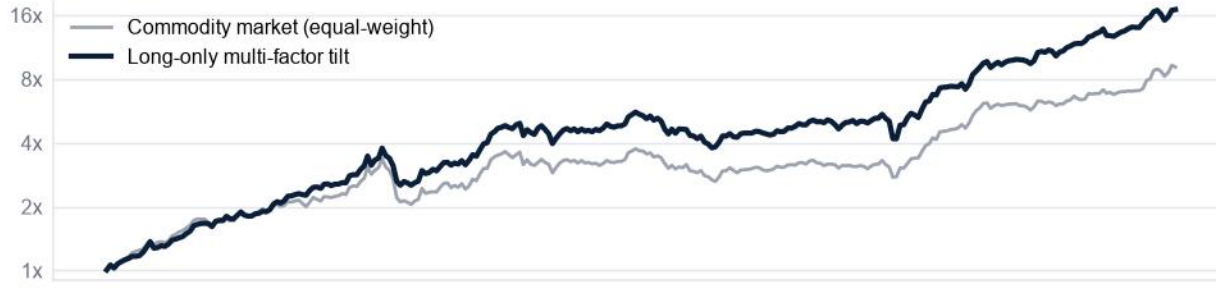
Gross monthly returns over overlapping months (AQR value and momentum to Jan 2025, TSMOM to May 2026). Both series scaled to 10% annual volatility for comparison. Source: AQR Capital Management, LLC, AQR Data Library.

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For long-only investors: a significant tilt

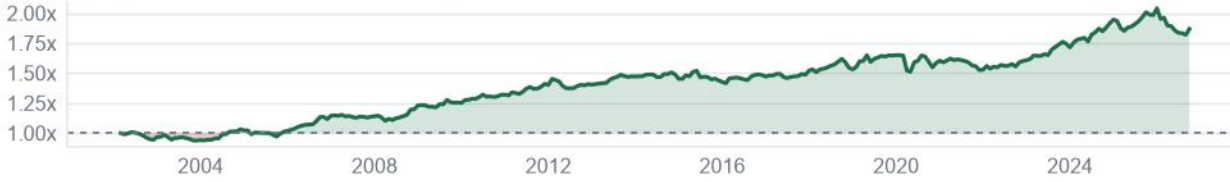
Equal parts of each factor's top-third basket beat the equal-weight commodity market by 2.7% a year (IR 0.53, t = 2.88) at similar volatility. No single tilt was significant alone; combining halved tracking error (10% to 5%).

Growth of $1, long-only (log scale)

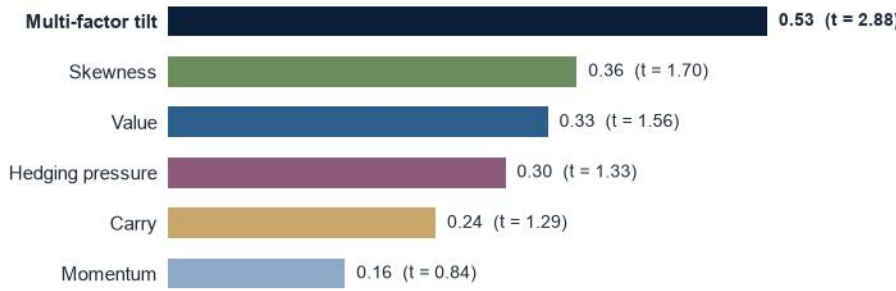


	Ann. return	Volatility	Sharpe	Max drawdown
Long-only multi-factor tilt	12.5%	14.2%	0.89	-33%
Commodity market (equal-weight)	9.9%	13.4%	0.74	-39%

Relative performance: tilt ÷ market (above 1 = ahead)



Information ratio vs the commodity market: single-factor tilts vs the combined tilt



Sample Feb 2002 to Sep 2026. Long-only tilt: equal-weighted top third of contracts on each factor, net of 5 bps rebalancing costs (roll costs are shared with the benchmark). Multi-factor tilt: equal parts of the five cross-sectional tilts.

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Method, robustness and caveats

How the results were built, which checks they passed, and what to keep in mind before relying on them.

HOW IT WAS BUILT

- **Universe**

  26 US futures across energy, metals, grains, softs and livestock, Feb 2002 to Sep 2026.

- **Data**

  Yahoo front-month prices, EIA curves, CFTC positioning, FRED macro. Free and public.

- **Returns**

  Roll dates from exchange expiry rules; roll gaps removed; WTI April 2020 handled explicitly.

- **Factors**

  Momentum, carry (12-month roll yield), value, skewness, hedging pressure, time-series momentum.

- **Portfolios**

  Monthly long-short terciles; trend sized by volatility; equal-risk blend of all six, fixed in advance. Costs: 5 bps per unit of turnover, including rolls.

ROBUSTNESS CHECKS PASSED

- ✓ **Independent replication**

  Correlation 0.75 to 0.79 with AQR's published factors.

- ✓ **Multiple testing**

  Combination t = 3.91 clears t > 3 and Bonferroni across 30 strategies (t > 3.14).

- ✓ **Extreme moves**

  Capping daily returns at ±20% leaves the combination Sharpe at 0.71.

- ✓ **Trading costs**

  Break-even cost 33 to 46 bps for the four strongest factors.

- ✓ **Construction**

  Results hold across weighting, sector-neutral and quintile variants.

CAVEATS

- ! **Optimistic levels**

  Sharpe ratios run above AQR's: front-month data holds contracts into delivery.

- ! **Declining edge**

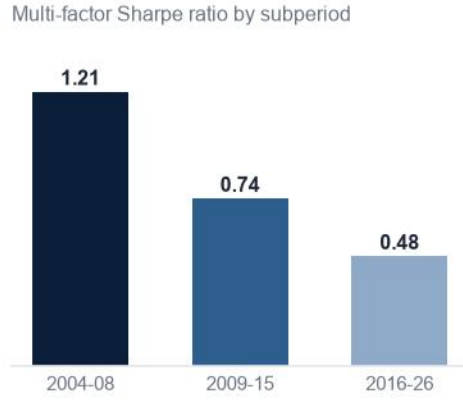
  Combination Sharpe fell from 1.21 to 0.74 and 0.48 across subperiods.

- ! **Weak factors**

  Momentum and carry earned little on their own; carry is limited by free data.

- ! **Backtest**

  Historical simulation, not live trading. Not investment advice.



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